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Long term flow change threatens invertebrate diversity in temporary streams: Evidence from an island

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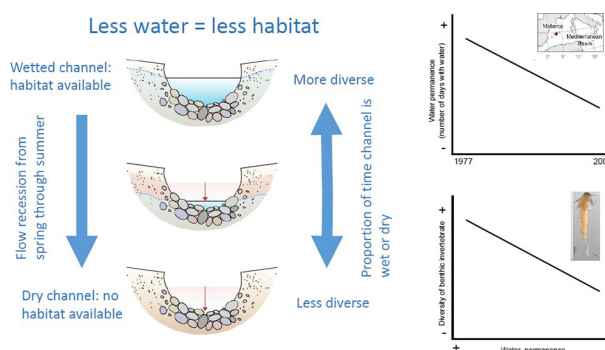
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HIGHLIGHTS

- Consistent downward trends in the number of days with water in spring and summer
- Propose 'number of flow days' as an ecologically-relevant metric in temporary streams
- Strongest trends when species are already constrained by limited water availability
- The potential impacts of long-term changes in temporary streams are acute on islands.

GRAPHICAL ABSTRACT



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ABSTRACT

Here we provide the first evidence of long term reductions in flow in temporary streams on the Mediterranean island of Mallorca and use a simple metric of the degree of water permanence (the number of days with water) to highlight the implications of flow change for aquatic invertebrate diversity. Analysis of a 33 year data set for 13 streams on the island yielded evidence of consistent downward trends in water permanence, particularly in spring and summer. Data from 27 relatively undisturbed mountain streams indicate that the diversity of benthic invertebrates in temporary streams across the island is directly related to water permanence. Streams with lower values of water permanence support few species overall and have less abundant invertebrate assemblages; the abundance and species richness of sensitive mayfly, stonefly and caddisfly taxonomic groups is also reduced in streams with lower water permanence. Although developed using spatial data, these flow-invertebrate relationships suggest that future reductions in water permanence may lead to reduced diversity. We argue that the 'number of days with water' is a simple but ecologically-relevant metric of water permanence that can be used effectively to monitor change in threatened temporary streams worldwide.

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1. Introduction

The world's temporary streams are recognised as supporting unique and important ecosystems (Hart and Finelli, 1999), but ones which are vulnerable to climate and land use change and to increasing human demand for water (Acuña et al., 2014). Temporary streams can be defined most simply as those which periodically cease to flow (MacDonough et al., 2011). Rather than being restricted to arid regions, such streams occur in most terrestrial biomes between 84° N and S, and represent a substantial proportion of the world's rivers (Tooth, 2000; Larned et al., 2010). There has been increasing interest in temporary streams over the last decade, mainly because the threats they face are now more widely recognised (Leigh et al., 2016). To date, much of the work on temporary stream ecosystems has concentrated on assessing their diversity relative to their permanent counterparts. Temporary streams typically support fewer species than permanent ones (Williams, 1996; Pardo and Álvarez, 2006; Rosenfeld et al., 2010) and a widespread concern is that, in the future, those species requiring perennial flow may disappear from streams which change from being permanent to temporary.

A related but fundamentally different question concerns how the diversity of temporary streams may be affected by increases in the duration of time that they are left dry. Organisms inhabiting temporary streams are adapted to periods without water (Williams, 2006) so flow cessation per se does not necessarily affect the survival of individuals or the persistence of whole populations. However, a sustained reduction in the number of days with water (i.e. reduced water permanence, sensu Rosenfeld et al. (2010)), may become critical if for example it prevents organisms from completing their life cycle (Boulton, 2003) or reduces stream seedbank composition (Stubbington et al., 2016). The effects of reductions in water permanence on species diversity in temporary streams remain poorly understood (Mokany et al., 2008). Most significantly from a conservation perspective, the rate at which species may disappear or experience reductions in population size as periods without water increase cannot currently be predicted with any certainty (Acuña et al., 2014). This is an important knowledge gap, because without such information it is difficult to assess the risks posed by hydrological change in temporary streams and, in turn, our ability to target conservation efforts to the most threatened streams or sites is compromised.

Issues of temporary stream conservation are particularly acute on islands. Island faunas are often less diverse than mainland ones as their geographic isolation limits colonisation by those mainland species which have poor dispersal ability (Pardo and Álvarez, 2006). However, islands are well known as having high levels of endemism (Whittaker and Fernandez-Palacios, 2007). Thus, islands often have unique and important faunas comprising a mixture of endemics and those ubiquitous species which have better dispersal abilities. Recent genetic and morphological analysis of aquatic insects on Macronesian islands has revealed the presence of endemic cryptic species (Rutschmann et al., 2014), suggesting that island faunas may actually be more diverse than typically recognised (Rutschmann et al., 2017).

Globally, Mediterranean-climate regions represent biodiversity hotspots (Cuttelod et al., 2008; Kier et al., 2009) and islands here are particularly important centres of endemism. Mallorca (Fig. 1) is one of ten biodiversity hotspots within the Mediterranean Basin (Médail and Quézel, 1999). Temporary streams with very limited human disturbance can still be found in mountainous parts of the island (García et al., 2014), and many of these support endemic invertebrates, including relics from previous humid Mesic periods (Malicky, 1992; Pardo and Zwick, 1993). There are concerns that, like in many other Mediterranean areas, streams on Mallorca are threatened by a combination of global (climate-driven) and local (landcover change and increased water demand) factors which alter groundwater resources and stream hydrological conditions (Candela et al., 2009; García et al., 2016). In particular, tourism-related water demand has led to over-exploitation of

the aquifers that are key elements for maintaining streamflow across the island (García and Servera, 2003). However, there is a lack of any empirical basis for predicting the likely effects of reductions in streamflow on aquatic invertebrate diversity. In this paper we assess long-term flow change in streams across Mallorca and use a simple transferable metric of water permanence to illustrate how invertebrate diversity may be affected by flow reductions.

2. Methods

2.1. Study area

Temporary streams are defined here as those which, for the whole or any part of their length, do not have water for the entire year. Thus, a temporary stream may have water for anything between zero and 364 days per year. This definition includes what are often termed 'ephemeral' or 'intermittent' streams - streams which flow only for limited periods following precipitation. Using this definition all streams on Mallorca (Fig. 1) can be considered temporary. The torrential character of the precipitation on the island, combined with permeable lithologies and relatively small drainage areas, causes Mallorcan streams to have highly seasonal flow regimes. Many dry up entirely for several months during the late spring, summer and earlier autumn as a result of reduced precipitation and cessation of discharges from spring sources. The exact period of time that their channels are dry varies from stream to stream and from year to year, reflecting local physiographic factors and inter-annual variation in climate.

2.2. Hydrological trends

Gauged streams in the Tramuntana region of Mallorca were used to assess hydrological trends. This is a mountainous area where human impact on landcover is limited. Streams were selected as (i) having long enough records to allow robust trend analysis (we set a minimum of 25 years for possible inclusion in the analysis, with no more than one year of missing data), and (ii) not having their flows regulated by dams or other major structures. In the end, 13 streams met these criteria. The streams ranged in size from 8 to 124 km², with the final time series consisting of data for the 13 streams for the period 1977–2009 inclusive (i.e. 33 years). Analyses were based on mean daily flow data by the Water Resources Department of the Balearic Islands Government.

The gauging stations differ in design (eight had water-level recorders and five staff gauges), but all are properly calibrated and the data used in the analysis have been officially validated by water authorities. Hydrological regimes of each stream over the study period were characterised using metrics designed to capture features of potential ecological relevance in temporary systems (Rosenfeld et al., 2010). The runoff coefficient was used to qualify basin discharge. This is a dimensionless parameter relating the total amount of runoff to the total amount of precipitation received. Specific discharge was used as a metric of flow magnitude; it expresses discharge as a function of basin area and so allows direct comparison of flow magnitude between streams of different size. Number of days with water (i.e. days when discharge is greater than zero) was used as a metric to represent the duration of time that streams were wet (and hence the temporal extent of dry periods).

For each stream, each flow metric was computed for the whole year, as well as for selected seasonal and monthly time intervals. Thus, in each stream we computed (i) mean annual runoff, (ii) mean annual specific discharge, (iii) annual number of days with water, (iv) number of days in spring with water, (v) number of days in summer with water, (vi) specific discharge in spring, and (vii) specific discharge in summer. Finally, we calculated number of days with water and specific discharge separately for the months of March–June inclusive for each stream. This gave a total of 20 statistical descriptors of flow that were assessed for trends over the 33 years. Rather than looking separately for trends in

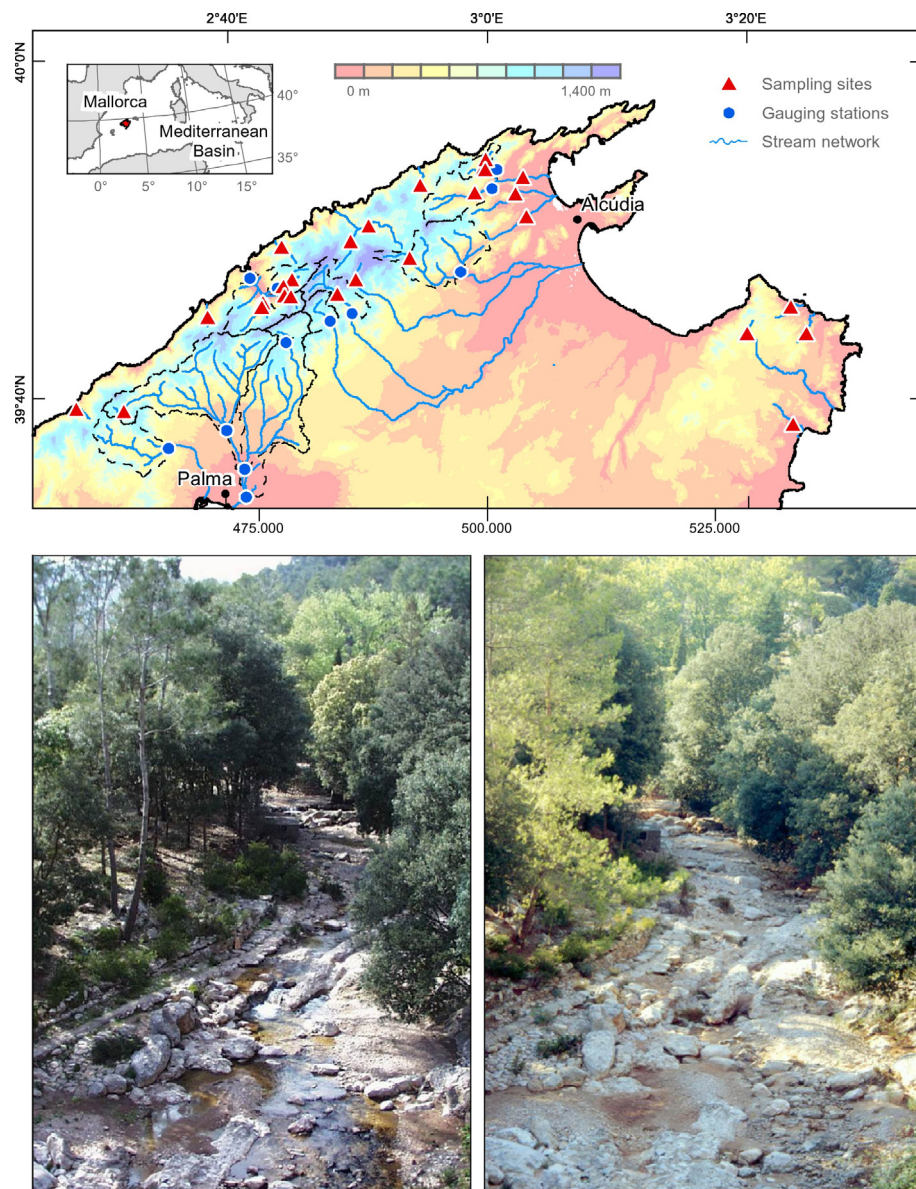


Fig. 1. Study area and streams. The upper panel shows the location of the gauging stations used for the hydrological analyses and the stream invertebrate sampling sites. The images in the lower panel show one of the study streams (Ternelles, northeast sampling point on map): the left image shows the stream under typical baseflow conditions, whereas in the right extensive parts of the channel appear desiccated during a dry period.

each stream, a Regional Kendall test (e.g. Helsel and Frans, 2006) was used to determine whether there was evidence of a common trend across all 13 streams for each of the 20 metrics.

2.3. Relationships between invertebrate diversity and water permanence

Analysis of macroinvertebrate diversity was based on part of a large data set (60 streams) collected between 2005 and 2008. This data set has already been used for a detailed study of benthic invertebrate community structure and ecological status across the Balearic islands (García et al., 2014). The present analysis was based on a subset of 27 of these streams, selected as being in quasi-pristine upland parts of the island, away from intensive agricultural areas where water courses are impacted by poor water quality, water abstractions and channelization. All 27 streams were considered by García et al. (2014) to have good ecological status. A single sampling site was located in each stream. Each site has continuous flow for a period exceeding 5 months each year. This flow criterion ensured the existence of fully developed communities rather than the species-poor ones, dominated by one or two taxa,

associated with sites which have flow for less than this time (Álvarez and Pardo, 2007; Delgado et al., 2012).

Samples were collected using a multi-habitat procedure with a kick-net of 500 μm -mesh size. Each site consisted of a 100 m long reach, with 20 1-min kick-samples collected and pooled to produce a single sample. Samples were distributed proportionally in the five most dominant habitats present (full details in García et al., 2014). Samples were preserved with ethanol (70%), transported to the laboratory and stored until being sorted and identified. Individuals were counted and identified under 40 $\times$ magnification (Olympus U-TV1X) microscope and a 5 $\times$ Nikon SMZ645. Identification was to mostly to either species or genus, although more difficult groups were left to higher levels (Diptera to either Family, Subfamily or tribe, Oligochaeta to Family and Hydrachnidia left as such). All samples were collected between November and June in 2005, 2006 and 2008. Each stream was sampled at least twice over this period, resulting in a total of 91 samples.

For each pooled sample 'taxon richness' (the total number of taxa present) and total invertebrate abundance (the sum abundance of all individuals of all species present in a sample) were used as metrics of

diversity. The species richness within taxonomic groups that are potentially sensitive to reductions in water permanence was also computed for each sample. The combined number of Ephemeroptera (E), Plecoptera (P) and Trichoptera (T) species present in each sample was used for this, as they are known to be sensitive to physical habitat conditions (Turley et al., 2014). These groups have high conservation importance in temporary streams on Mallorca as they consist of numerous rare species with very restricted distributions (Álvarez and Pardo, 2007). The abundance of each EPT species in each sample was also computed.

As very few of the 27 sites sampled for invertebrates are gauged, water permanence was quantified for each one using the approach of Rosenfeld et al. (2010). This involved determining whether or not the site was dry on each of the six times it was visited. For each site, water permanence was then quantified as the proportion of the six visits that water was present in the channel. This simple metric has the advantage over more complex ones that it captures the fundamental ecological constraint of whether water is present or not. Note that a water permanence value of 1 does not mean that a site has water permanently – rather, it simply means that water was present on all six sampling visits.

Relations between various metrics of invertebrate diversity (taxon richness, total abundance, and the total number of E, P and T species) were assessed using Generalised Estimating Equations (GEEs). As some species of E, P and T species were not present at all sites (e.g. several species occurred at <5 sites), the water permanence gradient was too short for meaningful analysis of the effects of flow on each species individually. Thus, relations between water permanence of abundance of the individual EPT species was assessed for 15 of the 21 members of this group. GEEs are linear models designed for use with repeat measurements or data that are correlated due, for example, to some spatial clustering of samples. GEEs were appropriate to our data because streams were sampled more than once and there was some geographic clustering (i.e. they were not distributed evenly across the island). Each stream was treated as a subject, while the repeat measurements from each stream were considered as within-subject effects. Analyses were applied to log-transformed values.

3. Results

3.1. Hydrological trends

There were significant downward trends in a number of the flow metrics over the study period (Table 1). At the annual timescale there were no significant trends in either runoff or discharge ($P = 0.45$ and 0.43 respectively). However, there was a consistent downward trend in the number of days each year that the streams held water ($P = 0.015$). At the seasonal timescale, there were significant downward trends in summer discharge ($P < 0.001$) and the number of days each

spring and summer the streams had water (both $P < 0.001$). Trends in discharge and the number of days with water were especially evident in March, April and June ($P = 0.006$, 0.005 and 0.004 respectively), while the March–June months all showed downwards trends in the number of days with water (all $P < 0.005$). Overall, the analyses provide evidence of long term hydrological changes in the streams.

3.2. Invertebrates

In total, 751,915 individuals belonging to 261 invertebrate taxa were identified. Six of the 21 EPT taxa recorded are endemic either to Mallorca or to the Mediterranean islands as a whole (Table 2). Of these important taxa, some were rather widespread (i.e. occurring in most of the streams) while others were recorded at very few sites.

There were significant positive relations between values of water permanence and both the number of taxa present and the total abundance of invertebrates (both $P < 0.001$; full details in Table 3; trends shown in Figs. 2a and b). Thus, although all the streams are temporary, those with water for a shorter period were less diverse. The number of EPT species present in the streams was also positively related to water permanence ($P < 0.001$; Table 3 and Fig. 2c). The abundance of 10 of the 14 EPT species assessed was positively related to water permanence ($P < 0.05$; Table 4). All the Ephemeroptera species, one of the two Plecoptera and five of the seven Trichoptera were less abundant in streams that held water for shorter periods.

4. Discussion

4.1. Threats to island streams

Islands are currently facing greater threats than mainland areas as a result of human activity, but because of their relatively small land areas and high levels of endemism they have been argued to offer high returns for species conservation efforts (Kier et al., 2009). Temporary streams occur worldwide and are recognised as being both sensitive and threatened by a range of factors influencing their flow regimes (Lake, 2003; Larned et al., 2010). Our study has shown that there has been a consistent reduction in water permanence in temporary streams on Mallorca over the last three decades. The analysis, undertaken of thirteen mountain streams, indicates that trends are evident at the annual timescale (total number of days with water each year) but are also occurring in spring and summer. Complementary data for 27 streams on the island provide statistical evidence that aquatic invertebrate diversity is directly related to water permanence – streams with water for fewer days support fewer species, less abundant communities and fewer species belonging to the sensitive Ephemeroptera, Plecoptera and Trichoptera. Our simple space–time substitution exercise therefore suggests that future reductions in water permanence may lead to

Table 1
Results of the Regional Kendall test. Formally significant trends ($P < 0.05$) are highlighted in bold.

		The tau correlation coefficient	S	Z	P value
Annual	Specific Discharge	0.026	190	0.78	0.436
	Days with water	–0.083	–569	–2.446	0.015
	Runoff coefficient	0.029	155	0.755	0.450
Spring	Days with water	–0.133	–910	–4.044	0.000
	Specific discharge	–0.048	–352	–1.498	0.134
Summer	Days with water	–0.152	–801	–4.754	0.000
	Specific discharge	–0.128	–788	–4.250	0.000
March	Days with water	–0.136	–935	–4.323	0.000
	Specific discharge	–0.086	–622	–2.739	0.006
April	Days with water	–0.13	–892	–4.250	0.000
	Specific discharge	–0.086	–622	–2.813	0.005
May	Days with water	–0.105	–668	–3.329	0.001
	Specific discharge	–0.036	–241	–1.138	0.255
June	Days with water	–0.112	–708	–4.302	0.000
	Specific discharge	–0.077	–508	–2.909	0.004

Table 2

Ecological and distributional information on all Ephemeroptera, Plecoptera and Trichoptera species recorded in the study.

	Number of streams recorded in	Seasons larvae present (1 = Autumn, 2 = Winter, 3 = Spring)	Adult flight times	Endemism
Ephemeroptera				
<i>Alainites muticus</i> Linnaeus, 1758	13	1,2,3	April	
<i>Baetis rhodani</i> Pictet, 1843	13	1,2,3	Not known	
<i>Cloeon simile</i> Eaton, 1870	10	1,2,3	Not known	
<i>Cloeon dipterum</i> Linnaeus, 1761	26	1,2,3	Not known	
<i>Caenis luctuosa</i> Burmeister, 1839	23	1,2,3	Not known	
<i>Thraulius bellus</i> (Eaton 1881)	2	2,3	Not known	
Plecoptera				
<i>Leuctra balearica</i> Pardo & Zwick, 1993	4	1,2,3	October, January	Yes
<i>Tyrrhenoleuctra antoninoi</i> Fochetti & Tierno de Figueroa, 2009	14	1,2,3	March, May	Yes
Trichoptera				
<i>Rhyacophila munda</i> McLachlan, 1862	9	1,2,3	April, May, October	
<i>Agapetus quadratus</i> Mosely 1930	5	2,3	May, October	Yes
<i>Stactobia</i> sp.	1	1,3	not known	
<i>Oxyethira</i> spp. (2 spp.)	6	1,2,3	April, May, October	
<i>Hydroptila</i> spp. (3 spp.)	24	1,2,3	May, October	
<i>Chimarra marginata</i> (Linnaeus, 1767)	2	1,2,3	April, May, October	
<i>Hydropsyche angustipennis mallorcana</i> Malicky, 1980	2	2,3	May, October	Yes
<i>Polycentropus kingi</i> McLachlan, 1881	13	1,2,3	May, October	
<i>Tinodes</i> spp. (<i>T. waeneri pollensa</i> Malicky, 1987; <i>T. mclachlani</i> Kimmins, 1966)	14	1,2,3	March, April, May, October	Yes
<i>Leptodrusus budtzi</i> (Ulmer, 1913)	3	2,3	May, October	Yes
<i>Micropterna fissa</i> McLachlan, 1875	1	1	March, August, October	
<i>Mesophylax aspersus</i> Rambur, 1842	17	1,2,3	March, April, July, December	
<i>Athripsodes braueri</i> Pictet, 1865	1	3	April, May, October	

changes in invertebrate diversity. The results have implications for other islands with finite water resources, notably those in the Eastern Mediterranean where reductions in precipitation have already been observed (Körner et al., 2005). Water permanence is simple to compute and intuitively obvious to interpret, and so provides a useful metric to help assess the threats faced by temporary streams; it provides a way to flag hydrological changes which may lead to ecological impoverishment.

4.2. Temporary stream ecosystems on Mallorca

Temporary streams on Mallorca have small basins that, during periods of heavy rainfall, experience torrential flows. These flows transform dry channels into shallow running waters during the spring season, at the end of which most aquatic species attain their highest densities (Álvarez and Pardo, 2009). Drying begins by the end of the spring and progresses during the summer months, although depending on channel characteristics deeper pools may persist and provide important over-summering refuges for aquatic insects (Boulton and Suter, 1986). Like others, habitat availability in temporary streams on Mallorca is mainly constrained by the absence of water and associated chemical changes (i.e. higher temperatures and lower dissolved oxygen at low flow; Williams, 1996). The larvae of many species are adapted to dryness, and are frequently reliant on mountain spring sources or

hyporheic refugia to allow them to persist over the summer period (Pardo and Álvarez, 2006). Aerial adults can undergo a summer diapause (aestivation) in nearby caves, where they wait for the autumn water resumption to lay their eggs (Salavert et al., 2011). Reductions in the permanence of water will reduce the availability of potentially usable (wetted) habitat as well as, critically, the distribution and connectivity of wetted habitat areas (Larned et al., 2010). Increasing pressures from agriculture and tourism not only imply greater water demand, but may also increase the risk of water pollution. This is evident in other streams on Mallorca (those not included in the present study) which fail to meet good ecological status criteria (García et al., 2014). Water quality issues have also been reported from the island of Madeira (Hughes, 2006). At present the implications of combined flow and water quality stressors for invertebrate community composition, phenologies and population persistence on Mallorca remain unclear.

The long term hydrological trends on Mallorca are unequivocal. Precipitation is the main driver for streamflow reduction, although the increase in temperature and land-cover changes also play a significant role in the decreasing of flows on the island (García et al., 2016). Such trends may be exacerbated in the future by increasing water demands from agriculture and tourism (Pascual et al., 2015). Precipitation is set to decrease in the coming decades in subtropical latitudes, particularly in the Mediterranean basin where droughts are anticipated to become longer and more frequent; e.g. global models project decreases in mean runoff in Mediterranean rivers of up to 30% (Schewe et al., 2014). All of these factors suggest that the trends evident in the 33 years of data presented here are likely to continue into the future.

Due to different tolerances, dispersal abilities, habitat requirements and life cycles, not all invertebrate species are likely to be affected to the same degree by reductions in water permanence. For example, the trichopteran *Mesophylax aspersus* is univoltine, with larvae needing five to six months to develop (Salavert et al., 2011). Consequently, reductions in water permanence to less than five months may have major implications for the distribution and abundance of this species. Conversely, species with better dispersal abilities or those with flexible

Table 3

Results of models fit to water permanence and invertebrate diversity metrics using Generalised Estimating Equations. Goodness of fit values computed using full log-quasi-likelihood functions.

	Wald Chi ² value	P value	Goodness of fit
Species richness	289.0	<0.001	10.7
Total abundance	23.6	<0.001	23.6
Number of EPT species	596.9	<0.001	14.9

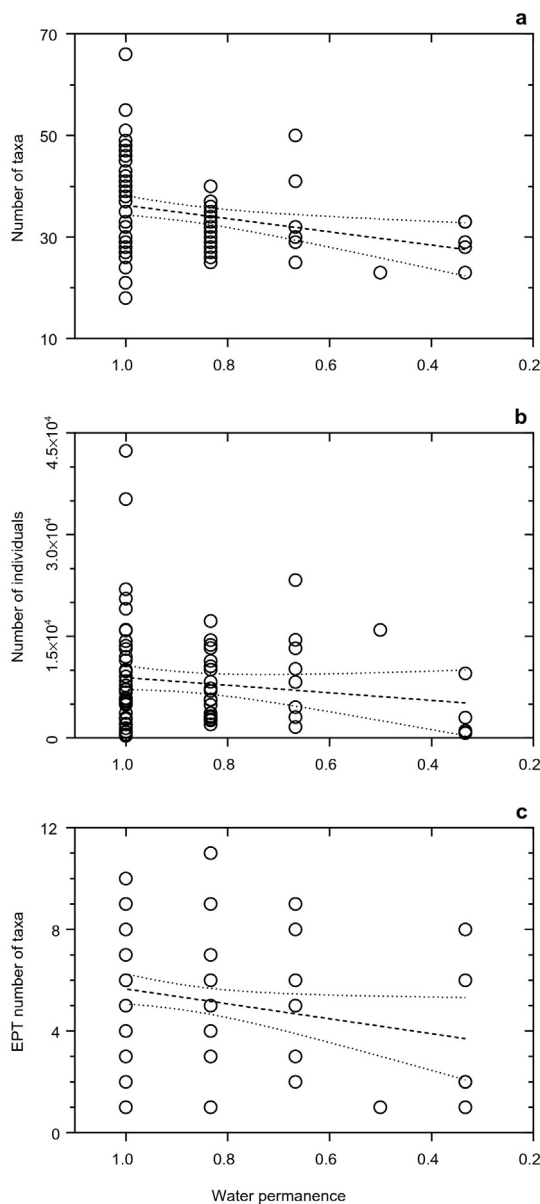


Fig. 2. (a) Invertebrate taxon richness, (b) total invertebrate abundance, and (c) the number of mayfly (Ephemeroptera), stonefly (Plecoptera) and caddisfly (Trichoptera) taxa (i.e. number of EPT taxa). Note that GEE models were fit to log transformed data, but to aid visual comparisons the raw values are shown here. Fitted lines, along with 95% confidence intervals, are superimposed over the raw data values.

life cycles (e.g. *Baetis* which may have either a polyvoltine or univoltine life cycle) may be less vulnerable to flow reductions.

It is notable that for our flow metrics the strongest trends occurred in the ecologically important spring and summer periods, when species are already constrained by limited water availability. Nevertheless, we have no grounds for thinking that there have been widespread extinctions on the island. For example, during our sampling campaigns we recorded all the Trichoptera genera and species listed by Malicky more than three decades ago (Malicky, 1992). Set against this are issues related to cryptic species. The species richness values reported here reflects current taxonomy, which is based on morphological assessments to delineate species. However, as found for other islands (Rutschmann et al., 2014; Rutschmann et al., 2017), there may be cryptic endemic species on Mallorca whose presence could be revealed by more detailed studies using genetic techniques. Thus, not only might the species richness values reported here be underestimates of the true diversity of

Table 4

Results of models fit to water permanence and species abundance values using Generalised Estimating Equations. Goodness of fit was computed using full log-quasi-likelihood functions. Note that only species occurring at >4 sites were analysed.

	Wald Chi ² value	P value	Goodness of fit
Ephemeroptera			
<i>A. muticus</i>	14.4	0.006	119.2
<i>B. rhodani</i>	19.8	0.001	149.1
<i>C. simile</i>	13.5	0.010	34.7
<i>C. dipterum</i>	46.3	<0.001	146.9
<i>C. luctuosa</i>	45.7	<0.001	120.9
Plecoptera			
<i>L. balearica</i>	4.2	0.086	56.7
<i>T. antoninnoi</i>	37.3	<0.001	85.8
Trichoptera			
<i>Rh. munda</i>	10.9	0.028	43.9
<i>A. quadratus</i>	7.2	0.066	36.3
<i>Oxyethira</i> spp.	4.8	0.190	27.1
<i>Hydroptila</i> spp.	11.4	0.022	126.6
<i>P. kingi</i>	25.2	<0.001	50.7
<i>Tinodes</i> spp.	20.2	<0.001	66.9
<i>M. aspersus</i>	38.7	<0.001	58.5

Mallorca's invertebrate fauna, but we may be appreciably understating the levels of endemism. These points underscore the need for conservation and careful water management on the island.

Multiple competing demands for water create major governance challenges worldwide (WRI, 2015). Traditional top-down approaches to water policy design have lost political legitimacy, and are being replaced progressively by more inclusive, bottom-up processes (Akhmouch and Clavreul, 2016). The degree of attention being given nowadays to expanding stakeholder involvement underscores how effective such engagement has proven to be in achieving sustainable water management objectives, and many frameworks now exist to help facilitate and formalise the engagement process (Lacroix and Megdal, 2016). In the case of temporary streams, stakeholder engagement is needed both to promote awareness of their importance and develop appropriate local policy. Engagement is especially critical on Mallorca because the small land-holdings, frequently old and artisanal water infrastructure, multiple land ownership and complex property boundaries mean that policies have to be built around specific local circumstances. Most particularly, local solutions are needed to the widespread overexploitation of the aquifers which help sustain surface streamflow for long enough to allow species to complete their life cycles. Tools already exist to aid stakeholder engagement in groundwater management (Eden et al., 2016) and these provide a foundation for sound water governance on the island.

5. Conclusions

Climate modelling studies and predictions of water demand suggest that in the coming decades hydrological stress will continue to affect temporary streams worldwide (Zhang et al., 2014). We therefore echo calls made recently by Acuña et al. (2014) and Datry et al. (2014) for monitoring, conservation and management to understand and address the challenges faced by temporary streams. The ecological impacts of long-term hydrological changes are particularly acute on islands such as Mallorca because of the endemism associated with their isolation from mainland areas (Palmer et al., 1999).

Reduced flows impact the quantity, quality, distribution and connectivity of wetted habitat available in temporary streams. What we have been able to show is that, despite the complex and interactive habitat changes associated with flow reductions, invertebrate density corresponds to a simple measure of water permanence (the number of days or the proportion of time that water is present). Thus, water permanence can be used as part of ongoing monitoring and assessment of the hydro-ecological status of temporary streams. Until more is

known about the ecology and habitat requirements of temporary stream invertebrates and especially the possibility of cryptic species on islands such as Mallorca, adoption of precautionary flow thresholds based on water permanence are warranted.

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